

Question Paper

Exam Date & Time: 21-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Digital Forensics [IT_3129]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A company discovers that an employee has been secretly modifying financial records for personal gain. Identify the type of traditional computer crime involved and describe how it fits under computer crime classification. (5)
- A)
- Course Outcome: "CO1" / Cognitive Level: "Remember"
- B) A hacker gains access to a user's bank credentials and later uses them to create a fake account for fraudulent transactions. Differentiate between identity theft and identity fraud in this context. (5)
- Course Outcome: "CO1" / Cognitive Level: "Understand"
- 2) In a cybercrime investigation, you are assigned to create an exact copy of a suspect's hard drive without altering the original data. Define forensic duplication and explain the types of duplication techniques commonly used in such investigations. (5)
- A)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- B) A cybersecurity analyst is assigned to collect digital evidence from a compromised workstation. Demonstrate how forensic technology and system tools can be applied to acquire and preserve digital evidence without altering its integrity. (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) During a cyber investigation, a forensic examiner is required to process a compromised Windows system. Apply the key steps involved in securing the scene, documenting findings, and collecting digital evidence systematically. (5)
- A)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Compare the functionalities of EnCase and FTK Imager in performing disk imaging and evidence validation during digital investigations. (5)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) In a cybercrime case where physical access is restricted, evaluate the key steps and challenges involved in performing remote acquisition while preserving evidence authenticity. (5)
- A)
- Course Outcome: "CO4" / Cognitive Level: "Analysis"
- B) Describe the significance of mobile device forensics and outline the challenges investigators face (5)

when extracting data from smartphones.

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5) How footprinting techniques are used to gather essential information about a target system before an attack. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Examine how enumeration reveals system details such as user accounts and shared resources that can be exploited by attackers. (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	1	5(10%)
2	Understand	3	15(30%)
3	Application	3	15(30%)
4	Analysis	3	15(30%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)